HTML and DOM, Model -View-Controller, Container, Data Objects and Business Objects, Persistent Object, Object Serialization using XML, Web Service

Objects and the Internet - Ajax, Object-Based Scripting Languages: JSON and Python

Object Oriented Design Principles - Single Responsibility Principle, Open/Close Principle, Liskov Substitution Principal, Interface Segregation Principle and Dependency Inversion Principle; Introduction to Component Based Design, Design Patterns and Code Smells.

Recommended Texts:

1. M. Weisfeld, *The Object-Oriented Thought Process*, 5th ed. Addison-Wesley Professional, 2019
2. P. Deitel and H. M. Deitel, *Java: How to Program*, 9th ed. Pearson College Div, 2011
3. H. Schildt and D. Coward, *Java: The Complete Reference*, 13th ed. McGraw-Hill, 2024

CSE 4304 **Data Structures Lab** **Credit: 1.50**

Sessional based on CSE 4303.

CSE 4308 **Database Management Systems Lab** **Credit: 1.00**

Sessional works based on CSE 4307.

SWE 4302 **Object Oriented Concepts II Lab** **Credit: 1.50**

Sessional works based on SWE 4301.

SWE 4304 **Software Project Lab I** **Credit: 1.50**

Each student will be assigned a single complete software project individually. The size of the projects will be medium. It will focus on the application of the different features of programming language. Student will be evaluated based on their software and problem solving effort.

Fourth Semester

Hum 4441 **Engineering Ethics** **Credit: 3.00**

Introduction to Engineering ethics and professionalism: What is engineering ethics? Why study engineering ethics? Responsible Professionals, Professions, and Corporations, The Origins of Ethical Thought, Ethics and the Law;

Moral Reasoning and Codes of Ethics: Ethical decision-making strategies, Ethical dilemmas, Codes of ethics, Case studies;

Moral Frameworks for Engineering Ethics: Ethical theories, Personal commitments and professional life;

Ethical Problem-Solving Techniques: Analysis of Issues in Ethical Problems, An Application of Problem-Solving Methods;

Engineering as Social Experimentation: Engineering as Experimentation, Engineers as Responsible Experimenters;

Risk, Safety, and Accidents: Assessment of safety and risk, Design considerations, uncertainty, Risk-benefit analysis, safe-exit and fail safe systems;

Engineer's Responsibilities and Rights: Employee/employer rights and responsibilities, Confidentiality and conflict of interest, Whistleblowing, Case studies on whistleblowing;

Honesty and Research Integrity: Truthfulness, Trustworthiness, Research Integrity, Protecting Research Subjects;

Computer Ethics: The Internet and Free Speech, Power Relationships, Property, Privacy, Additional Issues;

Environmental Ethics: Engineering, ecology, economics, Sustainable development, Ethical frameworks;

Global Issues: Multinational corporations, globalization of engineering, Technology transfer, appropriate technology;

Cautious Optimism and Moral Leadership: Cautious optimism as a technology development attitude, Moral leadership in engineering.

Recommended Texts:

1. C. B. Fleddermann, *Engineering Ethics*, 4th ed. Pearson, 2012
2. R. Schinzinger and M. W. Martin, *Introduction to Engineering Ethics*, 2nd ed. McGraw-Hill Education, 2009